

UGTS-KC 3.6.1 – BEA

Course-Corrected Synthetic Cell-Complex Edition

Boundary – Equivalence – Alignment

Formal integration

The source representations are modeled as synthetic one-dimensional cell complexes with explicit cycle annotations. Whitespace is converted into zero-metric hinge loops and then traded into bridge pathways while preserving Euler characteristic. Four remaining cycle generators are immersed into the coordinate circles of a synthetic four-torus. The exact XOR witness is constrained to an even-weight subspace, and the semantic residual to 19 becomes an exact signed-distance field in an evaluator-induced quotient metric.

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Revision status: course-corrected superseding edition of version 3.6.1

Source evidence is distilled without historical or dialect escalation.

Abstract

The attached source document combines four technical motifs: a four-loop text invariant with token splits, zero-width pathways, an ASCII XOR alignment, and a one-dimensional residual to the value 19. The earlier BEA report interpreted the first motif through ordinary font rendering. This revision supersedes that interpretation and follows the requested course correction: the source is represented by a *synthetic* finite one-dimensional CW-complex under an explicit profile.

The construction is exact. A representation with k tokens and four intrinsic cycles receives $k - 1$ auxiliary whitespace loops. Its augmented signature is $(\beta_0, \beta_1) = (k, 4 + k - 1)$, so $\chi = \beta_0 - \beta_1 = -3$ regardless of splitting. A Space-Hole Trading operation rewires each whitespace loop into a zero-metric bridge to the next token. Every rewire reduces both Betti numbers by one and leaves χ unchanged. Thus no neg moat changes from $(3, 6, -3)$ to $(1, 4, -3)$, matching negentien.

After collapsing the canonical backbone tree, the four independent cycles form a wedge of four circles. They map to the four coordinate circles of $\mathbb{T}_{\text{syn}}^4 = (\mathbb{R}/\mathbb{Z})^4$, and the induced first-homology matrix is I_4 . The exact 11-byte ASCII XOR delta has Hamming weight 38 and therefore lies in the even-weight subspace of $\mathbb{F}_2^{11 \times 8}$, preserving global parity. A fixed XOR translation is bijective, so Shannon entropy of a random fixed-width code variable is preserved, although the two fixed codewords have different Hamming weights. Finally, the evaluator quotient metric makes $F_{19} = \nu - 19$ an exact signed-distance field to the semantic zero set.

COURSE-CORRECTED DECISION

This document does not recover the earlier planar-font result. It defines a new, explicit synthetic profile in which the requested invariants are true by construction and verifiable by the bundled reference implementation. The scope label is part of every definition and certificate.

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1 Course correction and evidence boundary

1.1 What is being revised

The previous 3.6.1 BEA edition measured ordinary rasterized text and therefore found profile-dependent connected-component and counter counts. That was not the requested object. The present edition treats the source strings as named synthetic cell complexes. It supersedes the prior BEA interpretation while retaining the 3.6 literal definition registry, content hashes, dependency validation and query-first event architecture.

1.2 What the source contributes

The relevant source mechanisms are limited to the following page-level content:

- pages 1–2: two named representations, a claimed four-loop relation, leet forms and a scalar residual to 19;
- pages 3–4: the exact left-aligned, NUL-extended ASCII XOR table and repeated masks;
- page 5: the 1D cell-complex hinge, zero-width pathways and parity-hinge language.

The source PDF is registered by SHA-256:

7dc5894928bb3aa815824d35720a4a37a556fcc202c5a0f655dce46804f1f586

1.3 No contextual escalation

The strings are used as literal source labels. This revision makes no claim about a standard pronunciation rule, regional dialect, historical uniqueness or national singularity. It also does not infer physical energy, heat or material topology from the construction.

PROFILE BOUND

The exact claim is: *inside profile bea361:synthetic-cell-complex-v2, the definitions, maps, invariants and certificate below are internally exact.* The profile is not an assertion about ordinary font glyphs or Euclidean screen geometry.

2 Profile and literal representations

FORMAL DEFINITION

The course-corrected representation profile is

$$P_{\text{BEA}} = (\Sigma, N, \tau, A, w, \mathbb{T}_{\text{syn}}^4, \nu, 19),$$

where N is NFC/lowercase/single-space normalization, τ is a finite provenance-preserving transducer, A supplies explicit cycle annotations, w assigns zero metric weight to whitespace hinge cells, $\mathbb{T}_{\text{syn}}^4$ is the four-dimensional synthetic torus target and ν is a finite real-valued evaluator.

The shipped literal records are:

ID	Text	Value	Intrinsic cycle positions
repr:negentien	negentien	19	(1, 2, 3, 7)
repr:no-neg-moat	no neg moat	19	(1, 5, 8, 9)

Positions are zero-based indices in the normalized source strings. They are literal profile data, not font-derived counters.

The finite transducer is:

$$a \mapsto 4, \quad e \mapsto 3, \quad i \mapsto 1, \quad o \mapsto 0,$$

which yields:

```
negentien  -> n3g3nt13n
no neg moat -> n0 n3g m04t
```

Each output cell retains its input index and input symbol. This allows the cycle annotations to survive the transduction without reinterpreting the visual shapes of 3, 0 or 4.

3 Synthetic one-dimensional cell complex

3.1 Token paths and intrinsic cycles

Each token of m symbols becomes a path with m one-cells and $m+1$ vertices. Distinct tokens begin as distinct connected components. Every annotated cycle position adds one loop one-cell at its symbol anchor. Let h be the intrinsic cycle count; here $h = 4$ for both representations.

3.2 Whitespace loops

Every separator between adjacent tokens contributes one *space loop*: a loop edge (u, u) at the terminal vertex of the left token, tagged with the entry vertex v of the next token. Its metric weight is zero, but it remains a cell in the topology.

For k tokens and h intrinsic cycles, the augmented graph has

$$\beta_0 = k, \quad \beta_1 = h + (k - 1), \quad \chi = \beta_0 - \beta_1 = 1 - h.$$

The augmented Euler characteristic is therefore independent of the token split.

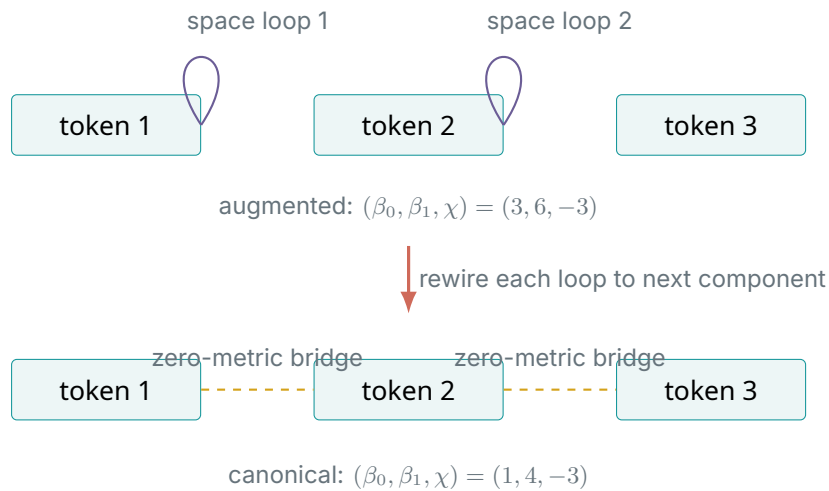


Figure 1: Whitespace is represented by auxiliary loops and then traded into zero-metric bridges. The two auxiliary cycles disappear exactly when the three components become one.

4 Space-Hole Trading Lemma

SPACE-HOLE TRADING LEMMA

Let X be a finite graph with a loop edge $e = (u, u)$ and let v lie in a connected component different from the component containing u . Replace e by the bridge $e' = (u, v)$ while leaving every vertex and every other edge unchanged. Then

$$\Delta\beta_0 = -1, \quad \Delta\beta_1 = -1, \quad \Delta\chi = 0.$$

PROOF / INVARIANT

The edge replacement leaves V and E fixed, so $V - E$ is fixed. The new bridge joins exactly two components, hence $\beta'_0 = \beta_0 - 1$. For a finite graph,

$$\beta_1 = E - V + \beta_0,$$

so $\beta'_1 = \beta_1 - 1$. Therefore

$$\chi' = \beta'_0 - \beta'_1 = (\beta_0 - 1) - (\beta_1 - 1) = \chi.$$

Iterating once per separator gives

$$(k, h + k - 1) \mapsto (1, h), \quad \chi = 1 - h \quad \text{at every stage.}$$

This is the exact meaning of “space-hole trading” in the revised profile: a separator loop is cut and opened into a bridge, so a cycle and a connected-component boundary disappear together.

4.1 Worked pair

EXACT METRICS

Representation and stage	V	E	(β_0, β_1)	χ
negentien, augmented	10	13	(1, 4)	−3
negentien, canonical	10	13	(1, 4)	−3
no neg moat, augmented	12	15	(3, 6)	−3
no neg moat, after trade 1	12	15	(2, 5)	−3
no neg moat, after trade 2	12	15	(1, 4)	−3

The source requires no separator trade. The target uses two. The canonical complexes have the same Betti signature and the same Euler characteristic under the declared synthetic topology.

5 Abstract four-torus immersing

5.1 Tree-collapse quotient

Let X^c be either canonical complex. It is connected with $\beta_1 = 4$. Choose a spanning tree T containing all symbol-backbone and zero-metric bridge edges. Collapsing T gives

$$X^c/T \cong \bigvee_{j=1}^4 S^1.$$

The four surviving loop edges provide an ordered cycle basis c_1, c_2, c_3, c_4 .

5.2 Synthetic torus target

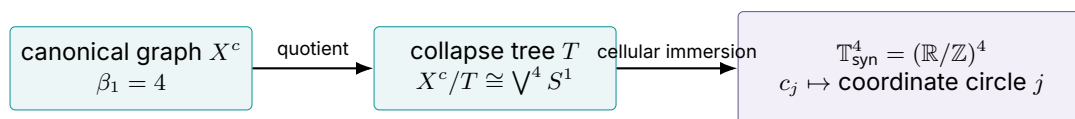
Define

$$\mathbb{T}_{\text{syn}}^4 = (\mathbb{R}/\mathbb{Z})^4.$$

The cellular immersion sends the j th cycle to the j th coordinate circle:

$$\iota(c_j(t)) = (0, \dots, 0, t, 0, \dots, 0) \pmod{1}.$$

The wedge point maps to the torus origin.



$$H_1(\iota) = I_4, \text{ rank } 4$$

Figure 2: The immersion is attached to the four-cycle quotient. It is not a claim that the text complex is the full four-torus.

The induced first-homology map is

$$H_1(\iota) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \text{rank } H_1(\iota) = 4.$$

PROFILE BOUND

The image is the union of four coordinate circles inside $\mathbb{T}_{\text{syn}}^4$, not the full four-dimensional manifold. The exact statement is a cellular immersion after tree collapse and an injective rank-four map on first homology.

6 Bitwise parity and entropy constraint

6.1 Exact alignment witness

The transduced byte strings are encoded in ASCII, aligned at the left and extended with NUL bytes to width 11. Let

$$X, Y \in \mathbb{F}_2^{11 \times 8}, \quad \Delta = X \oplus Y.$$

The source table is reproduced by the following exact rows:

i	source	source bits	Δ_i	delta bits	target
0	n	01101110	0x00	00000000	n
1	3	00110011	0x03	00000011	0
2	g	01100111	0x47	01000111	SPACE
3	3	00110011	0x5d	01011101	n
4	n	01101110	0x5d	01011101	3
5	t	01110100	0x13	00010011	g
6	1	00110001	0x11	00010001	SPACE
7	3	00110011	0x5e	01011110	m
8	n	01101110	0x5e	01011110	0
9	NUL	00000000	0x34	00110100	4
10	NUL	00000000	0x74	01110100	t

The delta bytes are:

00 03 47 5d 5d 13 11 5e 5e 34 74

The repeated rows are $\Delta_3 = \Delta_4 = 0x5d$ and $\Delta_7 = \Delta_8 = 0x5e$.

6.2 Even-weight subspace

Define

$$\mathcal{E}_n = \ker p, \quad p(\Delta) = \sum_{i=0}^{n-1} \sum_{j=0}^7 \Delta_{ij} \pmod{2}.$$

For $n > 0$,

$$\dim \mathcal{E}_n = 8n - 1.$$

The shipped witness has $n = 11$ and therefore lies in an 87-dimensional even-weight subspace. Adding the two disjoint row-equality constraints contributes 16 independent linear equations, leaving

$$88 - 16 - 1 = 71$$

dimensions in the symmetric subspace used by this witness.

6.3 Parity theorem

Let global parity be

$$\pi(X) = \sum_{i,j} X_{ij} \pmod{2}.$$

Since $Y = X + \Delta$ in $\mathbb{F}_2$,

$$\pi(Y) = \pi(X) + p(\Delta).$$

Thus $\Delta \in \mathcal{E}_n$ forces parity preservation.

EXACT METRICS

Metric	Exact value
aligned width	11 bytes
source Hamming weight	39
target Hamming weight	37
delta Hamming weight	38
source global parity	1
target global parity	1
even-subspace dimension	87
paired-row subspace dimension	71

6.4 Bounded entropy statement

For a fixed delta, the affine translation

$$\tau_{\Delta} : Z \mapsto Z \oplus \Delta$$

is an involutive bijection of the fixed-width code space. Therefore, for any random variable Z on that space,

$$H(Z \oplus \Delta) = H(Z).$$

Even delta weight is not needed for bijectivity; it additionally guarantees that each parity coset maps to itself.

PROFILE BOUND

The source and target fixed strings have different Hamming weights. The entropy result concerns a probability distribution under a bijective relabeling. It is not physical entropy, heat, energy or bit-count conservation.

7 Semantic-boundary SDF integration**7.1 Evaluator quotient**

Let R_P be the representation set and let

$$\nu : R_P \rightarrow \mathbb{R}$$

be the declared profile evaluator. Define

$$r \sim r' \iff \nu(r) = \nu(r').$$

The quotient $Q = R_P / \sim$ is identified with $\text{im } \nu$ and carries the metric

$$d_{\nu}([r], [r']) = |\nu(r) - \nu(r')|.$$

7.2 Pullback to spatial domains

Let Ω be the disjoint union of the cell-complex spatial domains and let $\lambda : \Omega \rightarrow R_P$ label every point by its representation. Pull back the metric:

$$\hat{d}_{\nu}(p, q) = |\nu(\lambda(p)) - \nu(\lambda(q))|.$$

This is a pseudometric on Ω : points in the same semantic fiber can be spatially distinct yet have distance zero. It becomes a genuine metric after quotienting by zero distance.

7.3 Exact signed distance to 19

Define

$$F_{19}(p) = \nu(\lambda(p)) - 19, \quad Z_{19} = F_{19}^{-1}(0).$$

When Z_{19} is non-empty,

$$|F_{19}(p)| = \inf_{z \in Z_{19}} \hat{d}_{\nu}(p, z).$$

Therefore F_{19} is an exact signed-distance field under the evaluator-induced metric.

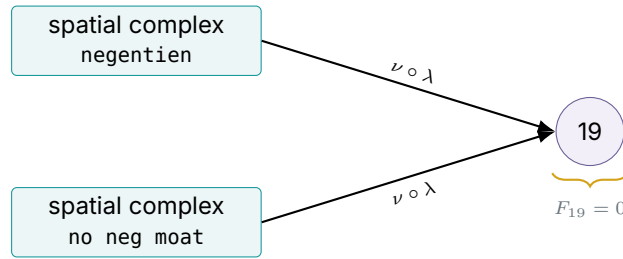


Figure 3: Distinct spatial complexes collapse to the same semantic quotient coordinate. The signed residual is exact in that metric.

For both shipped representations, $\nu = 19$, so $F_{19} = 0$ on every point of both spatial domains. A finite guard is

$$g_\varepsilon(r) = |\nu(r) - 19| - \varepsilon.$$

PROFILE BOUND

This is not the Euclidean signed distance to visible text. It is a spatially indexed field whose metric is induced by the scalar evaluator. The metric declaration is what makes the distance identity exact.

8 Integrated BEA certificate and substrate order

8.1 Certificate object

The course-corrected certificate is

$$C_{\text{BEA}}^{\text{syn}} = (P, r_s, r_t, X_s^+, X_t^+, H_s, H_t, X_s^c, X_t^c, \iota_s, \iota_t, \Delta, \mathcal{E}_n, \nu, F_{19}, N).$$

It records the profile, augmented and canonical complexes, every trade record, the torus maps, binary witness, parity/entropy scope, semantic field and explicit non-claims.

The shipped certificate is valid when:

1. every definition hash verifies and every reference resolves;
2. both canonical signatures are $(1, 4, -3)$;
3. each separator trade changes (β_0, β_1) by $(-1, -1)$ and preserves χ ;
4. both induced homology matrices equal I_4 ;
5. $X \oplus \Delta = Y$ and $Y \oplus \Delta = X$;
6. $\text{wt}(\Delta) = 38$ is even and global parity is preserved;
7. both semantic residuals to 19 are zero;
8. the profile scope accompanies the claim.

The resulting claim level is profile-exact-synthetic-topology-equivalence.

8.2 Referential execution order

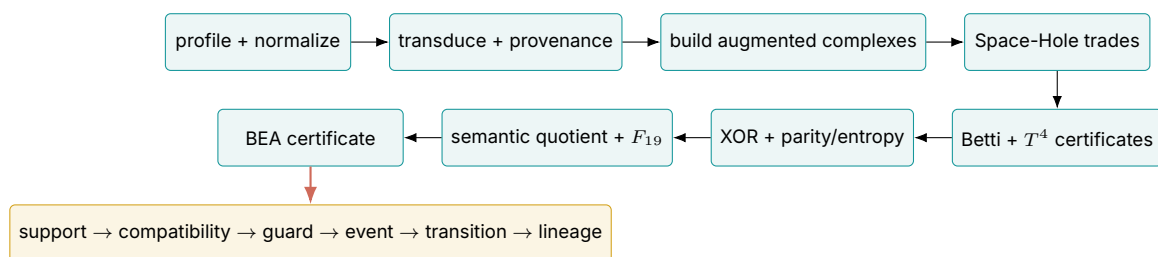


Figure 4: The BEA profile issues a scoped certificate and then hands control to the ordinary UGTS query-first event sequence.

9 BEA 3.6.1 delta operator catalog

The corrected namespace contains 30 atomic operators. Each entry includes a formal role, source/profile basis, disposition, implementation bound and validation method.

ID	Operator	Formal role and bound	Status
BEA361-D001	bea361.profile.synthetic-text-v2	$P=(\Sigma, N, \tau, L, w, T_4, \nu)$ fixes normalization, provenance transduction, explicit cycle annotations, zero-metric whitespace hinges, torus dimension 4 and target value 19. Bound: Valid only inside the named profile; not standard font topology.	PROFILE-AXIOM
BEA361-D002	bea361.normalize.canonical-space-v1	N lowercases, applies NFC and reduces every non-empty whitespace run to one separator while preserving separator count after normalization. Bound: Positional cycle annotations require unchanged layout.	ENGINEERING
BEA361-D003	bea361.transduce.leet-provenance-v1	τ maps $a \rightarrow 4$, $e \rightarrow 3$, $i \rightarrow 1$, $o \rightarrow 0$ while retaining each source index and source symbol as cell provenance. Bound: Finite versioned map; not a universal leet alphabet.	SOURCE-DERIVED
BEA361-D004	bea361.cell.token-path-v1	Each token of m symbols becomes a path with m edges and $m+1$ vertices; separate tokens begin as distinct components. Bound: One-dimensional CW model only.	PROFILE-AXIOM
BEA361-D005	bea361.cell.intrinsic-cycle-annotation-v1	Each declared cycle position adds one loop 1-cell at the symbol anchor. The source and target records each declare exactly four positions. Bound: Annotations are explicit data, not inferred glyph counters.	PROFILE-AXIOM
BEA361-D006	bea361.cell.whitespace-space-loop-v1	Each token boundary contributes one zero-metric loop edge at the right endpoint of the preceding token, tagged with the next token's entry vertex. Bound: Pseudometric weight zero does not remove the topological cell.	PROFILE-AXIOM
BEA361-D007	bea361.topology.betti-graph-v1	For finite graph X , β_0 is the component count and $\beta_1 = E - V + \beta_0$, with loop edges included. Bound: Graph/CW complexes only.	ENGINEERING
BEA361-D008	bea361.topology.euler-characteristic-v1	$\chi(X) = V - E = \beta_0 - \beta_1$. Bound: No planar-renderer interpretation is imported.	SOURCE-DERIVED
BEA361-D009	bea361.hinge.space-hole-trade-v1	Replace a loop edge (u, u) by a bridge (u, v) to a distinct component. V and E stay fixed; β_0 and β_1 both decrease by one; χ is conserved. Bound: Requires v in a different component and an actual loop edge.	PROFILE-AXIOM
BEA361-D010	bea361.hinge.zero-metric-bridge-v1	The traded separator becomes a bridge with metric weight 0. It joins components topologically while collapsing its metric length in the profile pseudometric. Bound: Weighted algorithms must accept a pseudometric.	PROFILE-AXIOM
BEA361-D011	bea361.hinge.iterated-space-collapse-v1	Apply the Space-Hole trade once per separator. For k tokens and h intrinsic cycles, $(\beta_0, \beta_1) = (k, h+k-1)$ becomes $(1, h)$ with $\chi = 1 - h$ throughout. Bound: Separator adjacency must be a tree/order chain.	ENGINEERING
BEA361-D012	bea361.torus.spanning-tree-collapse-v1	Collapse the canonical backbone/bridge spanning tree T , obtaining X/T homeomorphic to a wedge of β_1 circles. Bound: The quotient, not the uncollapsed tree, is the immersion domain.	ENGINEERING
BEA361-D013	bea361.torus.cycle-basis-four-v1	When $\beta_1 = 4$, order the four intrinsic loop edges by profile cycle index to obtain $c_1, \dots, c_4$ and $H_1(X/T; \mathbb{Z}) = \mathbb{Z}^4$. Bound: Reject if canonical β_1 is not exactly four.	PROFILE-AXIOM
BEA361-D014	bea361.torus.coordinate-circle-immersion-v1	Map $c_j(t)$ to the j th coordinate circle in $T_{\text{syn}}^4 = (R/\mathbb{Z})^4$; all four cycle generators have winding vectors e_j . Bound: Image is the union of four coordinate circles, not the full torus.	PROFILE-AXIOM
BEA361-D015	bea361.torus.homology-rank-certificate-v1	The induced H_1 matrix is I_4 and has rank 4. Bound: Certifies generator alignment, not homeomorphism to T_4 .	ENGINEERING
BEA361-D016	bea361.bits.ascii-fixed-width-v1	Encode the two transduced strings in 8-bit ASCII. Bound: ASCII only in the shipped witness.	SOURCE-DERIVED
BEA361-D017	bea361.bits.left-nul-alignment-v1	Embed both byte strings in length $n = \max(\text{lengths})$ by appending $0x00$ to the shorter source. Bound: Alignment and pad byte are part of the witness.	SOURCE-DERIVED
BEA361-D018	bea361.bits.xor-delta-matrix-v1	$\Delta = X \oplus Y$ in $\mathbb{F}_2^{n \times 8}$; applying the same Δ again inverts the affine translation. Bound: This is bitwise XOR, not a shift instruction.	SOURCE-DERIVED
BEA361-D019	bea361.bits.even-weight-subspace-v1	$E_n = \ker(p)$, $p(\Delta) = \sum_j \Delta_{ij} \bmod 2$. It has dimension $8n-1$ for $n > 0$. Bound: Guarantees parity class, not Hamming weight.	PROFILE-AXIOM

ID	Operator	Formal role and bound	Status
BEA361-D020	bea361.bits.symmetric-row-pairs-v1	The shipped affine subspace adds $\Delta_3 = \Delta_4$ and $\Delta_7 = \Delta_8$, reproducing masks 0x5d and 0x5e. Bound: Zero-based row indices; pair constraints are profile-specific.	ENGINEERING
BEA361-D021	bea361.bits.parity-coset-invariance-v1	For $Y = X + \Delta$, $\pi(Y) = \pi(X) + p(\Delta)$. Δ in E_n forces $\pi(Y) = \pi(X)$. Bound: One-bit global parity only.	ENGINEERING
BEA361-D022	bea361.bits.bijective-entropy-invariance-v1	$x \rightarrow x + \Delta$ is a bijection. For any random variable X on the fixed-width code space, $H(X + \Delta) = H(X)$; an even Δ additionally keeps parity cosets invariant. Bound: Does not assert equal empirical bit weight or physical entropy.	ENGINEERING
BEA361-D023	bea361.semantic.profile-evaluator-v1	ν maps each declared representation ID to a finite real value; the shipped source and target both map to 19. Bound: No external linguistic universality is inferred.	PROFILE-AXIOM
BEA361-D024	bea361.semantic.fiber-quotient-v1	$r \sim r'$ iff $\nu(r) = \nu(r')$. The quotient $Q = R / \sim$ is identified with $\text{im}(\nu)$ subset R . Bound: Evaluator/profile must be declared.	ENGINEERING
BEA361-D025	bea361.semantic.pullback-pseudometric-v1	For spatial samples p, q labeled by representations, $d_\nu(p, q) = \nu(\lambda(p)) - \nu(\lambda(q)) $. It is a pseudometric on space and a metric on Q . Bound: Distinct spatial points in one semantic fiber may have distance zero.	PROFILE-AXIOM
BEA361-D026	bea361.semantic.sdf19-v1	$F_{19}(p) = \nu(\lambda(p)) - 19$ and $Z_{19} = F_{19}^{-1}(0)$. Under d_ν , $ F_{19}(p) = \text{dist}_{d\nu}(p, Z_{19})$. Bound: Exact only in the semantic quotient/pullback metric, not Euclidean rendering space.	PROFILE-AXIOM
BEA361-D027	bea361.semantic.zero-set-guard-v1	$g_\epsilon(r) = \nu(r) - 19 - \epsilon$; commit when $g_\epsilon \leq 0$ under ordinary UGTS support and compatibility gates. Bound: epsilon must be finite and non-negative.	ENGINEERING
BEA361-D028	bea361.certificate.integrated-bea-v2	Record augmented/canonical Betti signatures, trade records, T4 homology certificates, XOR witness, parity/entropy scope, semantic quotient and explicit non-claims. Bound: Certificate validity is profile-exact and non-transferable without profile identity.	ENGINEERING
BEA361-D029	bea361.pipeline.query-first-handoff-v1	After the BEA certificate, continue support $\rightarrow$ compatibility $\rightarrow$ guard $\rightarrow$ verified event $\rightarrow$ transition $\rightarrow$ lineage. Bound: BEA does not replace support or compatibility.	ENGINEERING
BEA361-D030	bea361.lineage.course-correction-v1	Mark this package as the course-corrected 3.6.1 edition and retain the earlier 3.6 baseline as provenance, while superseding the prior planar-glyph BEA interpretation. Bound: Version remains 3.6.1; revision status must be explicit.	ENGINEERING

10 Reference implementation

10.1 Modules

Path	Responsibility
src/ugts36/bea_synthetic.py	Cell-complex builder, graph Betti evaluator, Space-Hole rewire, torus certificate, XOR subspace and semantic SDF.
src/ugts36/bea_runtime.py	Executes the literal pair pipeline from the content-addressed substrate.
examples/ugts_kc_3_6_1_bea_synthetic_example.json	The profile, operator definitions, literal source/target instances and execution schedule.
spec/ugts_kc_3_6_1_bea_synthetic.schema.json	Portable JSON Schema for the corrected substrate.
spec/bea_3_6_1_delta_operator_catalog.*	Machine-readable 30-operator catalog.
tests/test_bea361_synthetic.py	47 course-corrected tests.

10.2 Minimal use

```

from ugts36 import (
    SOURCE_ANNOTATION, TARGET_ANNOTATION,
    build_synthetic_bea_certificate,
)

certificate = build_synthetic_bea_certificate(
    SOURCE_ANNOTATION,
    TARGET_ANNOTATION,
)
assert certificate.valid
print(certificate.claim_level)
# profile-exact-synthetic-topology-equivalence

```

10.3 Reference bounds

- Finite ordered strings only; this is not a general CW-complex language.
- Cycle positions are explicit zero-based annotations and may not point to whitespace.
- The Space-Hole rewire requires a loop edge and a target in a distinct component.
- Zero-metric bridges create a pseudometric; algorithms must not assume positive edge lengths.
- The torus operator rejects any canonical graph with $\beta_1 \neq 4$.
- The torus certificate is symbolic; it does not numerically mesh a four-dimensional surface.
- The shipped binary witness is ASCII, left aligned, NUL padded and exactly 11 bytes.
- The entropy law is scoped to a fixed-width random variable under a fixed XOR translation.
- The evaluator must return finite real values and the semantic zero set must be non-empty.
- No physical hardware throughput or energy result is introduced by this revision.

11 Validation and exact results

11.1 Executable validation

The package runs the retained 36 baseline tests and 47 course-corrected BEA tests:

```
Ran 83 tests
```

```
OK
```

The corrected tests cover normalization, cycle annotations, general k -token Space-Hole iteration, graph formulas, torus rank and coordinate closure, exact XOR rows, subspace dimensions, parity, bijectivity, semantic metric laws, signed-distance identity, schema validation, content hashes, dependency order and end-to-end certificate execution.

11.2 Acceptance metrics

EXACT METRICS

Metric	Result	Acceptance condition
source canonical (β_0, β_1, χ)	$(1, 4, -3)$	exact
target augmented (β_0, β_1, χ)	$(3, 6, -3)$	exact
target canonical (β_0, β_1, χ)	$(1, 4, -3)$	exact after 2 trades
Euler preservation per trade	true	$\Delta\chi = 0$
torus homology rank	4	both representations
XOR roundtrip	true	both directions
delta weight/parity	38 / even	parity preserved
repeated mask pairs	$(3, 4), (7, 8)$	exact
semantic residuals	0, 0	target 19
JSON Schema errors	0	required
definition hash failures	0	required
reference/dependency failures	0	required
combined tests	83 passed	required

11.3 Kill criteria

The profile certificate must be rejected if any of the following occurs:

- a cycle annotation is absent, duplicated, out of range or placed on whitespace;
- a space loop points into its own connected component before the trade;
- a trade changes V or E , fails to reduce both Betti numbers or changes χ ;
- the canonical cycle rank is not exactly four;
- the homology matrix loses rank;
- the XOR witness does not roundtrip, has odd delta weight when parity preservation is claimed, or violates declared row symmetries;
- the semantic evaluator is missing, non-finite or has no zero-set representation at value 19;
- a result is exported without the profile ID and non-claims.

12 Revision ledger and closing statement

12.1 Superseded interpretation

The earlier 3.6.1 BEA report treated the source pair as ordinary rendered text and concluded that boundary features did not match. That result belonged to a different representation profile. This course-corrected edition does not erase that measurement; it declares it irrelevant to the requested synthetic object and supersedes it as the formal BEA 3.6.1 definition.

12.2 Retained UGTS architecture

The corrected BEA layer remains referential and content-addressed. It does not replace the core event calculus. After a valid BEA certificate is issued, the authoritative sequence remains:

support → compatibility → guard → verified event → transition → lineage.

One-bit values remain schema-bound fields; continuous state, uncertainty and identity stay separate.

COURSE-CORRECTED DECISION

UGTS-KC 3.6.1 BEA is now defined exactly as requested: a synthetic one-dimensional cell-complex profile with Euler-preserving Space-Hole trades, a four-cycle immersion into a synthetic T^4 , an even-weight parity-preserving XOR subspace with a bounded entropy theorem, and a semantic signed-distance field to 19 under an explicit quotient metric.

A Machine-readable artifacts

The accompanying ZIP includes the full editable source, 30-operator CSV/JSON catalog, JSON Schema, literal substrate, exact XOR rows, deterministic trace, source register, Python implementation, all tests, validation report and SHA-256 manifest. The raw uploaded source PDF and font files are not redistributed.

B Attribution record

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